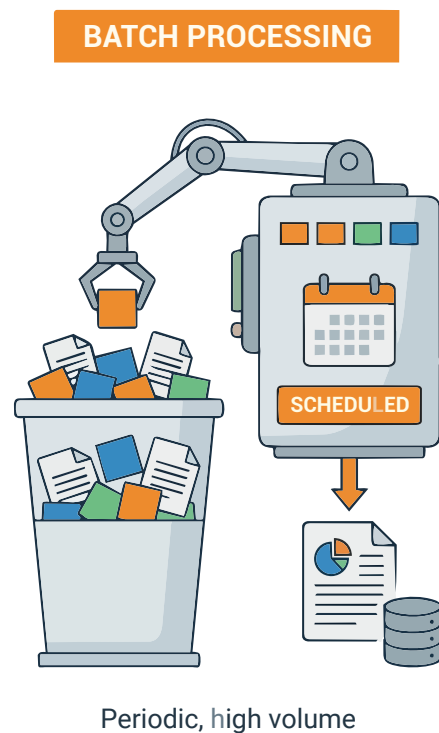
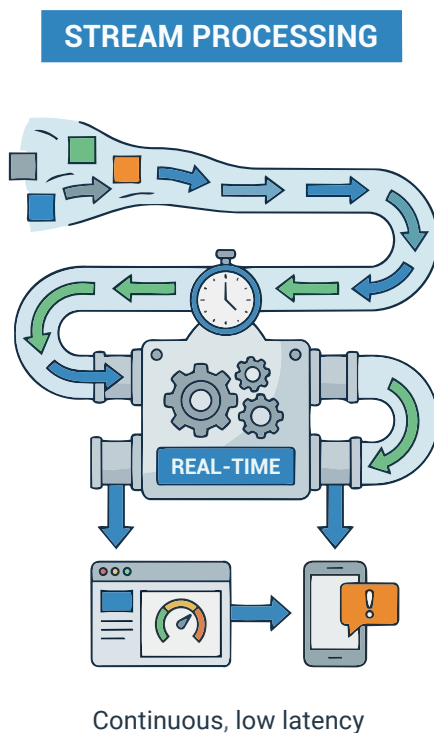


Stream vs Batch Processing: When to Choose What

Batch and streaming are two different ways of processing data, and both are relevant in today's world. Pick the approach that's right for you without falling for the hype that you must always do everything in real time.



A few years back I was running point on a data project that, in hindsight, should have been a lot simpler than it ended up being. We had a marketing team that wanted to see customer activity in their dashboard. Pretty standard stuff. Show me how many people signed up today. Show me which campaigns are converting right now. Show me when somebody high-value lands on the pricing page so I can email them while they're still warm.

We built it as a nightly batch job. Every night at 2 a.m., a Python script would pull data from a dozen sources, transform it, dump it into the warehouse, and refresh the dashboard. It worked great. For about six weeks.

Then the marketing director walked over to my desk one morning, slightly grumpy, holding her phone. "Why am I seeing yesterday's numbers? It's almost lunchtime." I tried to explain that the dashboard refreshed once a day. She didn't

really care about the explanation. She wanted to see customers landing on the pricing page now, not tomorrow morning. That was the whole point of the dashboard.

That conversation kicked off a six-month migration from a batch pipeline to a streaming one. We made a lot of mistakes along the way. We over-engineered some things. We under-engineered others. By the end of it, I had a pretty strong opinion about when streaming makes sense and when it doesn't, and which open source tools were genuinely worth the learning curve.

What does batch mean in practice?

Batch processing is one of the oldest and most widely used data processing methods because it remains highly effective. Data is collected over a fixed period—such as an hour, day, or week—and then processed in a single run. The job reads,